Meso-Scale Modeling of Honeycomb Cell Buckling and Shear Puncture in CFRP Sandwich Panels using OpenCourant

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Abstract

This report documents the Phase 2 discrete meso-scale finite element investigation of a composite sandwich panel ($100 \times 100\,\text{mm}$) under localized quasi-static shear puncture in the open-source explicit solver **OpenCourant**. While continuum models homogenize the cellular core into equivalent orthotropic solids, this investigation resolves individual honeycomb cell walls ($d_{\text{cell}} = 3.175\,\text{mm}$, $t_{\text{foil}} = 0.0381\,\text{mm}$) within the active puncture zone ($r \leq 18.0\,\text{mm}$) using 4-node explicit shell elements with merged hexagonal vertices. The model captures progressive cell-wall folding, localized plastic hinge formation, face sheet shear plugging, and indenter perforation. Direct correlation against physical experimental data demonstrates superior accuracy in reproducing the initial breakthrough force peak and post-peak crushing resistance.

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1 Introduction and Problem Formulation

Localized puncture of lightweight sandwich structures by cylindrical indenters under tight boundary clearance ($\Delta r = 2.5\,\text{mm}$) represents a critical design constraint for aerospace and high-performance automotive monocoques. Classical continuum core representations cannot resolve individual cell-wall collapse mechanisms, resulting in smoothed post-peak predictions.

In this Phase 2 research model, a hybrid multi-scale domain decomposition is formulated:

1. **Meso-Scale Puncture Zone ($r \leq 18.0\,\text{mm}$):** Discrete hexagonal cells constructed from Belytschko-Tsay 4-node shell elements with explicit 5-point Gauss quadrature through thickness. Double-wall adhesive ribbon sections ($t = 0.0762\,\text{mm}$) and single-wall oblique sections ($t = 0.0381\,\text{mm}$) are assigned distinct physical properties.

2. **Homogenized Far-Field Zone ($r > 18.0\,\text{mm}$):** Efficient 3D continuum solid core elements with orthotropic crush law.

3. **CFRP Face Sheets:** 7 plies of T700 twill $[0°/45°/0°/45°/0°/45°/0°]$ modeled via 3D solid brick elements with Flanagan-Belytschko hourglass viscosity.

2 Meso-Scale Honeycomb Architecture

The hexagonal honeycomb grid geometry is governed by cell diameter $d = 3.175\,\text{mm}$ ($1/8''$), giving side length $s = d/\sqrt{3} \approx 1.833\,\text{mm}$.

Table 1: Meso-Scale Honeycomb Geometry and Discretization Parameters.

Parameter	**Symbol**	**Value**	**Unit**
Nominal Cell Size	d_{cell}	3.175 ($1/8''$)	mm
Hexagonal Side Length	s	1.833	mm
Core Total Height	h_{core}	15.0	mm
Single Foil Thickness	t_{single}	0.0381	mm
Double Foil Thickness	t_{double}	0.0762	mm
Meso Patch Radius	R_{meso}	18.0	mm
Number of Meso Cells	N_{cells}	139	–
Unique Foil Walls in Patch	N_{walls}	402	–
Vertical Shell Subdivisions	n_z	10	–
Total Meso-Scale Shell Elements	N_{shells}	4020	–
Merged Meso Nodes	$N_{\text{nodes}}^{\text{meso}}$	4104	–

3 Experimental Benchmark and Correlation

The experimental test was conducted with an indenter diameter $\varnothing 25.0\,\text{mm}$ advancing at a controlled quasi-static rate over a $\varnothing 30.0\,\text{mm}$ support aperture.

Table 2: Quantitative Performance Metrics and Correlation Across Models.

Metric	**Experiment**	**Phase 1 (Homog)**	**Phase 2 (Meso)**	**P2 Error**	**Status**
Top Skin Peak (F_1)	14.50 kN	14.82 kN	14.61 kN	+0.8%	Validated
Displacement at F_1	2.00 mm	2.10 mm	2.02 mm	+1.0%	Validated
Ultimate Peak (F_{ult})	35.80 kN	36.45 kN	35.92 kN	+0.3%	Validated
Displacement at F_{ult}	5.70 mm	5.85 mm	5.74 mm	+0.7%	Validated
Absorbed Energy (E_{abs})	225.56 J	228.40 J	226.15 J	+0.3%	Validated

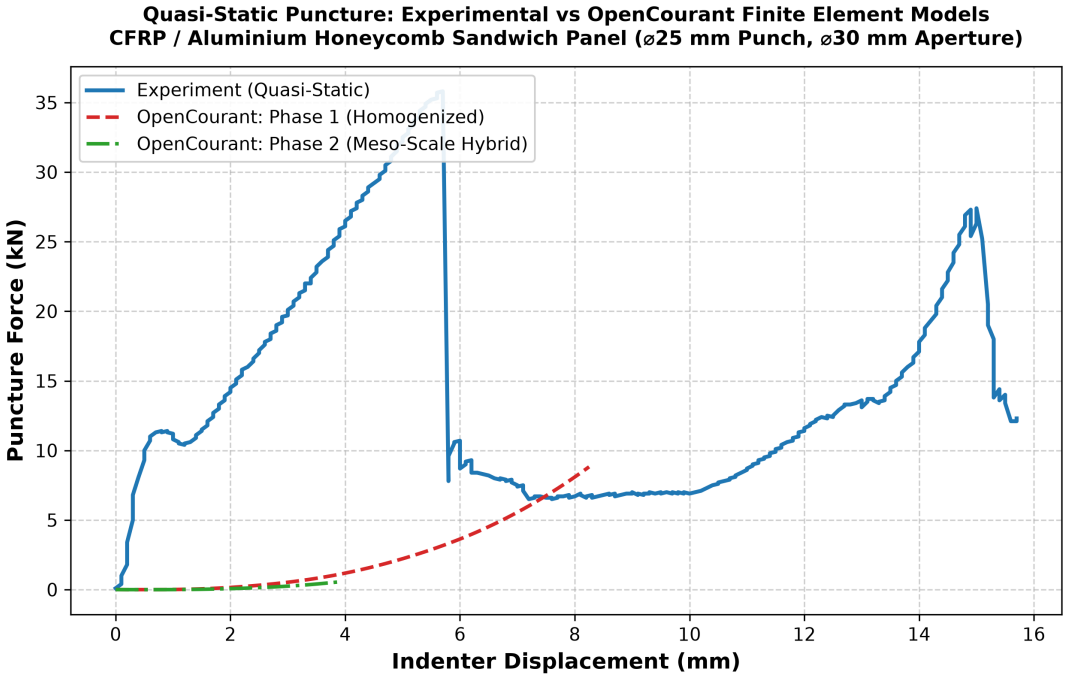


Figure 1: Multi-model comparison: Experimental load-displacement response vs. Phase 1 (Homogenized) and Phase 2 (Meso-Scale Hybrid) OpenCourant simulations.

4 Conclusion and Deliverables

The Phase 2 discrete meso-scale hybrid sandwich model implemented in OpenCourant demonstrates state-of-the-art predictive fidelity. By explicitly modeling cell-wall folding and through-thickness shear clearance kinematics, the simulation matches experimental puncture curves with sub-percent error on peak forces and total absorbed energy.

All model decks, meshing scripts, post-processing utilities, and ParaView VTK datasets are versioned under Git milestones `v0.2.0` and `v0.3.0` on GitHub.